



Curriculum Vitae

Erik Tabuchi Barczak

Personal Details

Name Erik Tabuchi Barczak
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Citizenship Poland, New Zealand

Work Experience

- Mar. 2017 - **Student Assistant**, Technical University of Darmstadt, Institute of Computer
Nov. 2018 Integrated Design (DiK).
Web application for the Space Factory 4.0 project. Feature-based recognition of branched sheet metal.
(C#, HTML/JavaScript)
- Jun. 2019 - **Systems Lead**, Sony, Hawk-Eye Innovations.
Feb. 2020 Goal Line Technology/Video-Assistant-Referee. On-site troubleshooting and remote support for hardware & software systems at live sports events.
- Feb. 2020 - **Software Developer**, MedCom GmbH, VeriSuite.
Nov. 2021 Computer vision for patient positioning during proton therapy, x-ray generator interface, GUI and multi-threading functionalities.
(C/C++, WinAPI, CUDA)
- Nov. 2021 - **Academic Researcher**, Karlsruhe Institute of Technology (KIT), Institute of
Nov. 2023 Industrial Information Technology (IIIT).
AgiProbot - Anomaly detection and 3D reconstruction using multi-sensor systems (2D & 3D cameras, robotic arm). Supervision of the digital signal processing lab course.
(C++, Python, ROS, PyTorch, GitLab CI/CD)
- Nov. 2023 - **Self-employed Software Developer**, N12 Tactics,
present Funded by Projektträger Jülich.
Design and development of cross-platform apps (iOS/Android, Windows/macOS) for tactical conception and video analysis in sports.
(Flutter/Dart, C++, Python)

Education

- 2007 - 2011 **High School Diploma**,
Northcote College, Auckland, New Zealand.
NCEA Merit Endorsement
- 2013 - 2016 **Bachelor's Degree**, Technical University of Darmstadt,
Computational Engineering (B.Sc.).
Grade: 2,73
- Oct. 2016 **Bachelor's Thesis**, Institute of Computer Integrated Design (DiK).
Development of a feature library for the web-based application for the modeling of higher
order branched sheet metal. Grade: 1,0
(JavaScript/HTML)
- 2016 - 2019 **Master's Degree**, Technical University of Darmstadt,
Computational Engineering (M.Sc.).
Grade: 1,86
- Jun. 2019 **Master's Thesis**, Institute of Computer Integrated Design (DiK).
A DT-VR-RC (Digital Twin, Virtual Reality, Remote Control) Process Chain for Assembly,
Integration, and Testing of Spacecraft Components on an In-Orbit Platform. Grade: 1,0
(C#, Unity3D)

Technologies & Tools

- Programming Languages C/C++, Dart, Python, C#, JavaScript/HTML.
- Frameworks Flutter, OpenCV, WinAPI, ROS, CUDA, PyTorch, Point Cloud Library, Open3D, GStreamer.
- Tools Git, GitLab CI/CD, Countersoft Gemini (ticket system).

Languages

- English Native speaker.
- German Business fluent.
C1 certificate 2012, DSH exam 2013 (TU Darmstadt)
- Portuguese Fluent.
Grew up bilingual
- Japanese Basic.
NCEA Level 1 and 2, Northcote College

Publications

- Dec. 2020 Incremental Improvement for Sub-optimal Euclidean TSP Paths Generated by
Traditional Heuristics.
- Jul. 2023 A Framework for Anomaly Classification and Segmentation in Remanufacturing
using Autoencoders and Simulated Data.
- Oct. 2023 European Machine Vision Forum: Primitive-based Automatic Digitization of
Mechanical Products.
- May 2024 6D Pose Estimation on Point Cloud Data through Prior Knowledge Integration:
A Case Study in Autonomous Disassembly.
- Dec. 2024 Are Benford's Law Based Detectors Effective for GAI Generated Images.